DEPI GRADUATION PROJECT

ResearchVoice AI

Speak the data. Keep the research moving.

Voice-assisted capture of research variables, with evidence checks, researcher approval and Excel export.

SYNTHETIC-ONLY RESEARCH PROTOTYPE



Five contributors · One instructor-approved integrated n8n workflow

The data entry burden repeats at every visit

Research observations move through several manual steps before they reach a clean spreadsheet.

01
Collect
Interview or review notes

→

02
Re-enter
Type into a CRF / form

→

03
Check
Find omissions and typos

→

04
Correct
Reconcile before analysis

Business problem Repeated manual entry costs attention and creates rechecking work. The time impact is not yet benchmarked.

A focused first user — the clinical researcher

The buyer is a pilot hypothesis until the team confirms a budget-owning decision-maker.

Primary user

Clinical researcher / physician-investigator collecting repeated study variables and follow-up records.

Beneficiaries

Research coordinators, study teams and data managers who receive more consistent approved records.

Potential buyer

Principal investigator, research unit, or hospital/university research administration — not yet confirmed.

Value hypothesis: less repetitive transcription, clearer exceptions, and researcher-controlled approval.

One integrated system, five accountable components

The instructor approved one connected n8n workflow rather than five separate workflow files.

C1

Study + schema

Approve variables

→

C2

Intake + safety

Voice / text screening

→

C3

AI extraction

Evidence-bound mapping

→

C4

QA + monitoring

Judge · alerts · LangSmith

→

C5

Review + export

Human confirmation → XLSX

Each component has one team owner [assign names]. All five team members explain their part in the pitch.

The researcher stays in control at every handoff

A spoken value is not accepted just because an AI model extracted it.

1	**Define study**	Approve fields and units
2	**Capture**	Synthetic Telegram text/voice
3	**Review transcript**	Correct words before AI
4	**Extract + judge**	Flag uncertain values
5	**Confirm**	Edit and approve all fields
6	**Export**	Deliver researcher-approved XLSX

NO AUTOMATIC CLINICAL DECISIONS

One synthetic follow-up from message to Excel

Use the current Telegram workflow; keep voice optional to reduce live-demo uncertainty.

Telegram demo input

“The patient is 54 years old and HbA1c is 6.2 percent.”

SYNTHETIC CASE · TRACE-DEMO

Age

54 years

HbA1c

6.2 percent

Researcher checks → confirms → XLSX

What the team has actually demonstrated

One documented Phase 4 synthetic scenario; not a validated performance benchmark.

1

Soft alert triggered

Ambiguous HbA1c variable

1

Approved XLSX delivered

After manual correction

4.6/5

Mean AI judge score

One uncalibrated evaluation

10.49 s

Judge processing duration

Judge portion only

The judge score is not clinical accuracy. No measured time savings, production reliability or financial ROI are claimed.

A useful failure: the model did not silently fill a mismatch

The voice transcript used an altered HbA1c field name; the workflow flagged it for researcher correction.



Overall: needs review. A soft alert prompted a manual correction before approved export.

Value will be measured — not assumed

The targets below are proposed; actual pilot KPI results are still pending.

KPI	CURRENT BASELINE	PROPOSED TARGET	ACTUAL
Preparation time / visit	8 min planning estimate	≤ 5 min	Not measured
Exact field agreement	Baseline pending	≥ 95%	Not measured
Correction rate	Baseline pending	≤ 15%	Not measured
Approved XLSX success	Baseline pending	≥ 95%	1 demo; no benchmark
Researcher ease score	Baseline pending	≥ 4/5	Not measured

No ROI or clinical benefit claim until the proposed baseline is checked in paired synthetic trials.

Human approval is a requirement, not a marketing feature

ResearchVoice AI is assistive research data capture — never diagnosis or prescribing.

IMPLEMENTED IN PROTOTYPE

- ✓ Study-approved variable schema
- ✓ Transcript confirmation and editable fields
- ✓ Missing/conflict warnings before export
- ✓ Explicit researcher approval and XLSX delivery
- ✓ QA, soft alerts and LangSmith checkpoint

LIMITATIONS / PILOT GATES

- ! PHI checks are best-effort; voice reaches Groq first
- ! n8n static storage is not a clinical database
- ! QA judge is not independently calibrated
- ! LangSmith logs a QA checkpoint, not every call
- ! IRB/privacy/security approval needed before real data

A structured pilot — before claims of impact

We are requesting evaluation access and feedback, not claiming an existing purchase or clinical deployment.

1	**Validate the problem**	20 researcher interviews (target), confirm buyer
2	**Run a controlled synthetic pilot**	10 proposed testers, 2 interested research teams (targets)
3	**Calibrate & measure**	Two human raters, KPIs, end-to-end cases
4	**Decide on readiness**	Security, privacy, IRB and durable-data plan

Our ask: permission for a researcher-led synthetic pilot, access to reviewers, and feedback from a potential buyer.

A better research record starts with a reviewed draft.

ResearchVoice AI

Questions • Safety • Workflow • Evidence • Pilot readiness

DEMO | SYNTHETIC INPUTS | HUMAN APPROVAL

Five presenters · Roles to assign before rehearsal